Get Started with Text Analysis in Azure

Microsoft Foundry / Azure Language delivers powerful NLP capabilities — sentiment analysis, key phrase extraction, entity recognition, and language detection — to turn raw text into structured, actionable insights.

PREREQUISITE: BASIC AZURE PORTAL KNOWLEDGE

Introduction to Text Analysis & NLP

Natural Language Processing (NLP) enables machines to understand, interpret, and respond to human language — extracting meaning and structure from unstructured text. The goal: convert raw text into actionable insights that improve decision-making.

Customer Feedback

Detect sentiment trends and improve experience

Healthcare

Extract symptoms, diagnoses, and medications

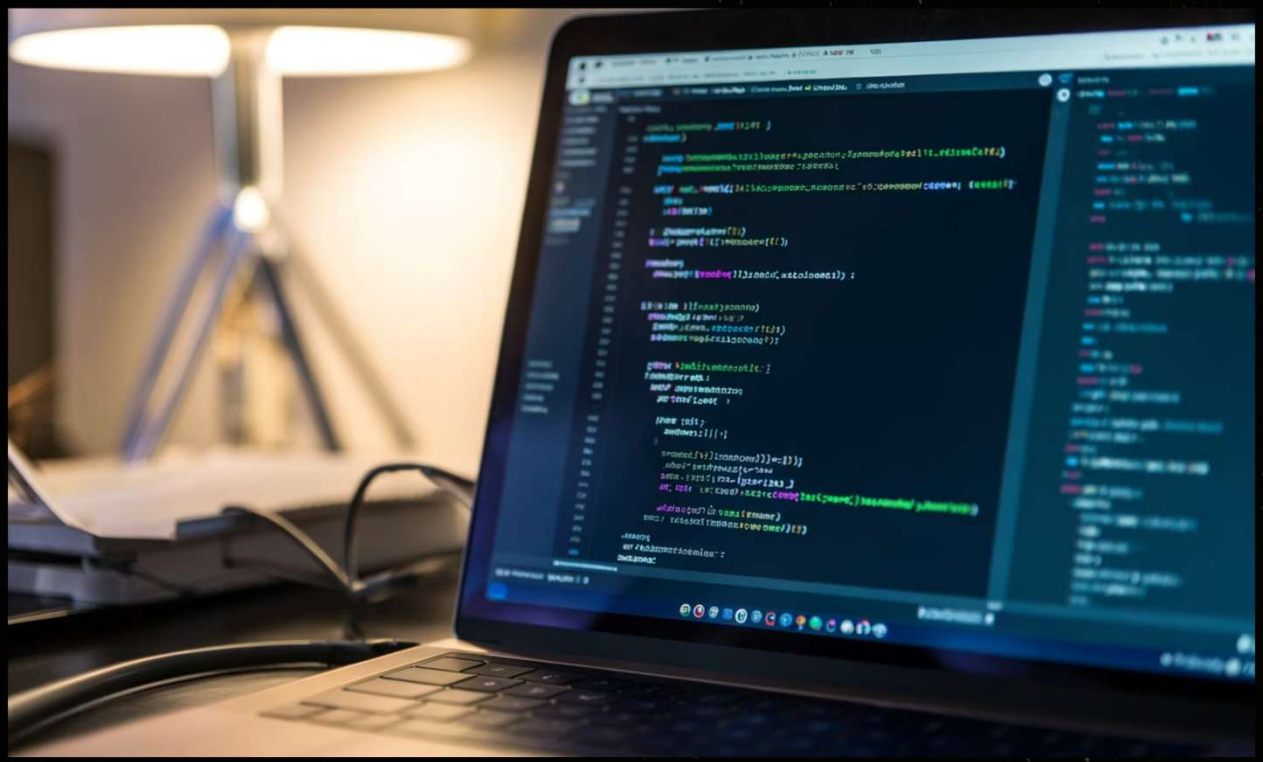
Finance

Process contracts and detect risks

Legal

Summarize documents and highlight clauses

Azure Language in Microsoft Foundry



Azure Language is a fully managed NLP service within Microsoft Foundry, offering pre-built models for common language tasks — no ML expertise required.

Key Phrase Extraction

Identifies main concepts and topics

Named Entity Recognition

Detects people, places, and organizations

Entity Linking

Connects entities to knowledge bases like Wikipedia

Language Detection

Identifies language and dialect automatically

Advanced Features & Foundry Playground

Sentiment Analysis

Scores text as positive, neutral, or negative with confidence metrics

Summarization

Generates extractive and abstractive summaries from long documents

PII Detection

Identifies and flags sensitive personal information in text

Foundry Playground

Test all features interactively — no coding required. Requires a Foundry resource and project.

The Playground accelerates prototyping for real-world scenarios: reviews, healthcare records, financial documents, and more.

Azure Language SDK Overview

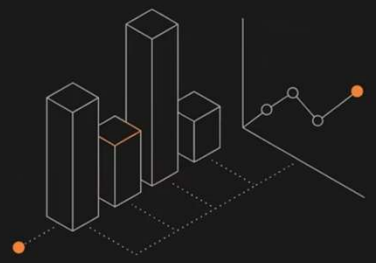
What It Is

The Azure Language SDK is a client library that integrates NLP features directly into your applications — no manual REST calls needed.

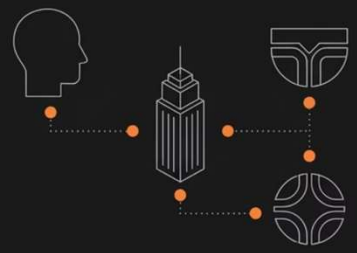
- Supports **Python, JavaScript, C#, and Java**
- Authenticates via endpoint + API key from Foundry
- Returns structured **JSON responses**

Key Concept

The **Client Object** handles authentication, sends requests, and parses responses — simplifying every interaction with the service.



Sentiment Analysis
Score text polarity.

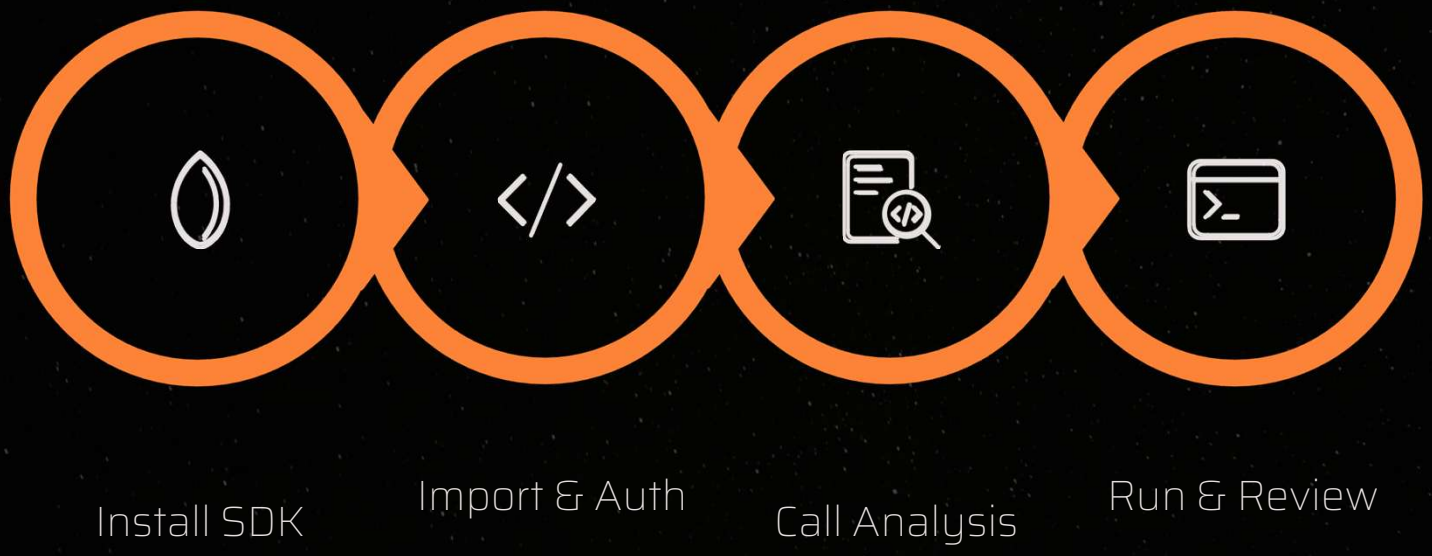


Entity Recognition
Detect names, places, orgs.

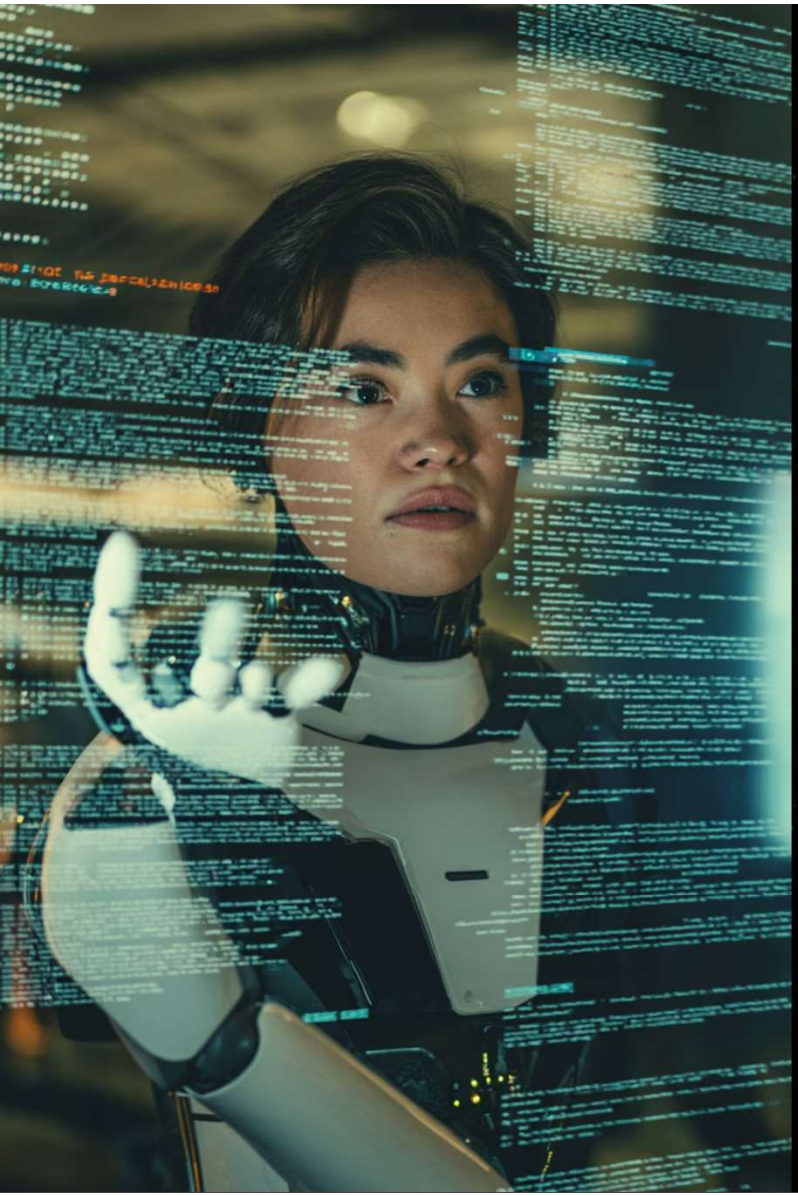
Key Phrase Extraction
Identify main concepts.

Language Detection
Identify language and dialect.

Using the Azure Language SDK in Python



With just a few lines of code, you can build AI-powered applications that automatically extract insights from text — sentiment, entities, key phrases, and summaries — all through a clean, consistent SDK interface.



Azure Language MCP Server

The **Model Context Protocol (MCP) Server** exposes Azure Language capabilities as managed tools that AI agents can call directly — no custom integration or separate resource required.

Add as a Tool

Register Azure Language in the Foundry Playground as an agent-accessible tool

Configure Connection

Link the MCP server to your existing Foundry resource — no extra setup

Trigger via Prompts

Use natural language prompts to execute sentiment, entity, and summarization tasks

Build Agentic AI

Agents perform structured, accurate text analysis as part of automated workflows

Summary: Text Analysis in Azure

What You've Learned

Azure Language transforms **unstructured text into structured, actionable insights** — enabling intelligent systems that understand, analyze, and act on language data at scale.

NLP Foundation

Sentiment, entities, key phrases, summarization, and language detection

SDK Integration

Embed text analysis programmatically into any application

MCP for Agents

Enable AI agents to perform language tasks within workflows

Next Step: Open Microsoft Foundry, create a Language resource, and explore the Playground to test text analysis on your own data.